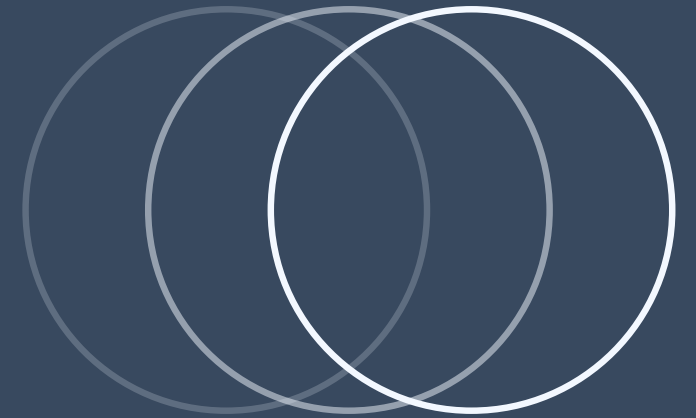


CSE 400



HANDOVER PROBABILITY ANALYSIS IN DRONE CELLULAR NETWORKS

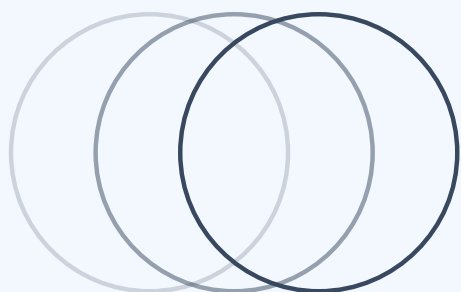


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PROJECT OBJECTIVE •

Derive probability distributions for random variables and compute $P[H(t)]$ analytically for SSM and DSM models.

- Understanding PDF/CDF formulas for all random variables
- Deriving nearest-neighbor distance distribution from PPP
- Generating simulation plots to validate theoretical distributions
- Explaining why DSM yields lower handover probability than SSM



PROJECT SYSTEM OVERVIEW

Mathematical Framework Flow:



1. PPP Theory

Void probability formula $\rightarrow$ Nearest-neighbor distribution

2. Distribution Assignment

Assign PDF/CDF to positions, directions, speeds

3. Temporal Evolution

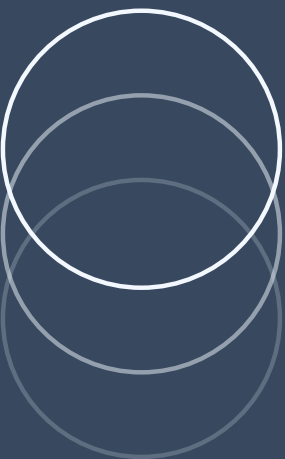
$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}t[\cos\theta, \sin\theta]$$

4. Integration

Condition on (r, θ) , integrate over all values

5. Handover Probability

$P[H(t)]$ from geometric arguments and simulation validation



SOURCES OF UNCERTAINTY

01

Spatial Randomness

PPP with density λ_0 gives void probability: $P[\text{no drone in } A] = \exp(-\lambda_0|A|)$

02

Directional Randomness

$\theta \sim U[0, 2\pi)$ creates isotropic movement patterns

03

Speed Randomness (DSM)

$V \sim \text{Rayleigh or Uniform}$ introduces heterogeneous velocities

04

Geometric Randomness

Minimum distance $u^*(t)$ evolves unpredictably due to combined uncertainties

KEY RANDOM VARIABLES WITH DISTRIBUTIONS



$\Phi D(0) \sim PPP(\lambda_0)$

Nearest neighbor: $f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2) \mid F(r) = 1 - \exp(-\pi\lambda_0 r^2)$

$\theta \sim U[0, 2\pi)$

$f(\theta) = 1/(2\pi) \mid F(\theta) = \theta/(2\pi)$

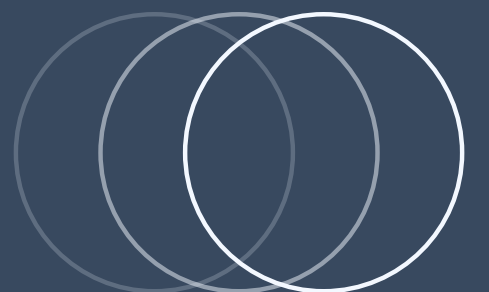


$V \sim \text{Rayleigh (DSM)}$

$f(v) = (v/\sigma^2)\exp(-v^2/2\sigma^2) \mid F(v) = 1 - \exp(-v^2/2\sigma^2)$

$u(t)^*$

Distance to serving drone - time-dependent, computable for SSM



PROBABILISTIC MODELS AND ASSUMPTIONS

01

PPP for Positions

- Tractable void probability formula
- Displacement theorem preserves PPP structure
- Models random deployment without clustering

02

Rayleigh for Speed

- Most drones near average speed
- Extreme speeds are rare
- Mathematically convenient

03

Uniform for Direction

- No preferred direction
- Ensures isotropy

PROBABILISTIC REASONING

Layer 1 - Spatial Statistics

PPP void probability $\rightarrow f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2)$

Layer 2 - Temporal Evolution

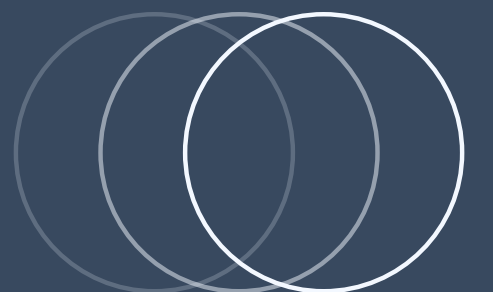
$x(t) = x_0 + vt[\cos\theta, \sin\theta]$

Layer 3 - Event Probability

Handover if new drone enters closer region

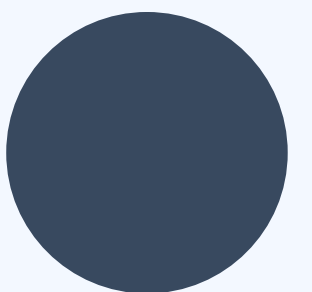
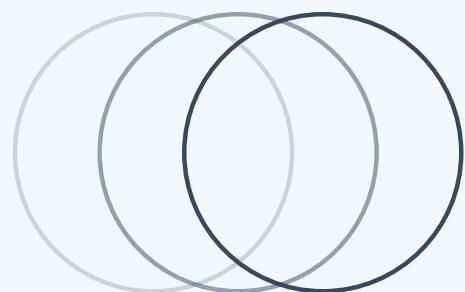
Layer 4 - Integration

Condition on (r, θ) , integrate with proper densities



OPERATIONALIZING - SIMULATION & VALIDATION

- **Monte Carlo Simulation:**
- **Step 1:** Generate $N \sim \text{Poisson}(\lambda_0 \times \text{area})$
- **Step 2:** Assign $\theta_i \sim U[0, 2\pi)$, $v_i \sim fV(v)$
- **Step 3:** Update $x_i(t) = x_i(0) + v_i t [\cos\theta_i, \sin\theta_i]$
- **Step 4:** Track handover events
- **Step 5:** $P[H(t)] = (\text{handover count}) / (\text{total trials})$
- Parameters: $\lambda_0 = 1 \text{ DBS/km}^2$, $v = 45 \text{ km/h}$, $t = 0-100\text{s}$



CURRENT UNDERSTANDING GAPS

Mathematical Details:

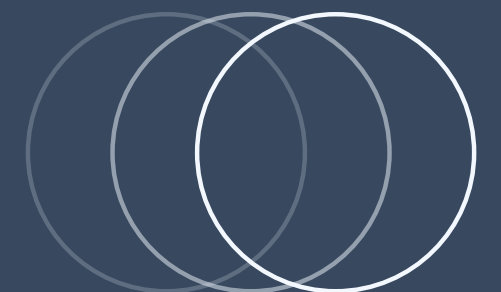
- Exact derivation of Lemma 3 density formula
- Numerical evaluation of Theorem 2 integrals
- Full proof of Lemma 2 (once-leaving property)

Conceptual Questions:

- Why exactly does $DSM < SSM$ intuitively?
- How tight is the DSM lower bound quantitatively?
- Sensitivity to distribution choice (Rayleigh vs Uniform)?

Extensions Needed:

- Different drone heights
- Random waypoint mobility
- Mobile users



PLANNED REFINEMENTS FOR M3

01 - Simulation Deep Dive

- Understand Monte Carlo convergence
- Study confidence intervals
- Validate PDF/CDF plots match theory

02 - Mathematical Validation

- Verify all distribution derivations
- Check formula interpretations
- Ensure parameter choices are justified
-

03 - Team Coordination

Everyone: Study simulation structure

Scribe: Document derivations

Reviewer: Verify analytical vs simulation match

GitHub: Organize plots and code

SUMMARY OF M2 UNDERSTANDING



M1: "PPP means random positions"

M2: PPP has void probability $\exp(-\lambda_0|A|)$ giving $f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2)$



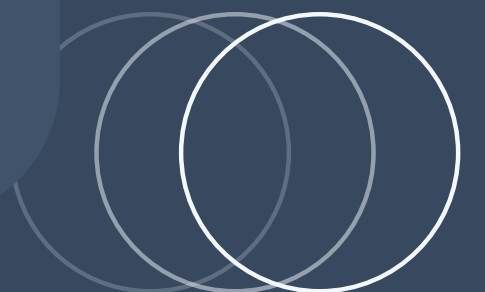
M1: "Theorem 1 equivalence is clever"

M2: Works via PPP translation invariance - relative motion is what matters



M1: "DSM can't get exact answer"

M2: Different speeds break similarity transformation; Jensen's inequality explains direction





THANK
YOU

